Cote 140-2

[scan 501]

thus by $(*)$ we have $u(l_i) \in L_i$ i.e. $u(L_i) \subset L_i$, but we have $\mathrm{Aut}(L_i) \simeq \widehat{\mathbb{Z}}^*$, and an endomorphism|automorphism of the|of a † $\widehat{\mathbb{Z}}$-module|modules free|free of rank 1 L_i is known when one knows it on an $nL_i \overset{\sim}{\leftarrow} L_i$. Thus $\forall i$ we have $u(l_i) = l_i$, thus $u = \mathrm{id}$, q.e.d. [p. 254]

In the case where π is not free, take a basis $(x_i, y_i)_{1 \leq i \leq g}$ – so that π is defined by the generating relation

$$\prod_1^g [x_i, y_i] = 1, \tag{3}$$

let Λ_i, Λ_i' be the closed subgroups generated by x_i, y_i. (They are moreover all conjugate under $\mathrm{Aut}(\pi)$…) I admit that these groups are $\simeq \widehat{\mathbb{Z}}$ (i.e. that the surjective homomorphisms $\widehat{\mathbb{Z}} \longrightarrow \Lambda_i$, $\widehat{\mathbb{Z}} \longrightarrow \Lambda_i'$ sending 1 to the x_i, y_i are injective – this † this is moreover trivial on passing to π_{ab}) and that we have, by analogy with (2), for every $n \in \mathbb{N}^*$

$$\mathrm{Centr}_\pi(\Lambda_i^n) = \Lambda_i, \qquad \mathrm{Centr}_\pi(\Lambda_i'^n) = \Lambda_i', \tag{4}$$

Uncertainty retained from cancelled passages.

140-2-s501-D001 L_i

140-2-s501-D004 should not be hard

[scan 502]

and we finish as above.

Remark The strongest faithfulness result in the direction of the present §, concerning actions of $\mathbb{\Gamma}$ and of its open subgroups, would be the following: if V is an anabelian algebraic curve over the finite extension k of $\mathbb{Q}$, with the algebraic closure $\bar{k}$, then not only should the outer action of $\mathcal{E}_k^{\bar{k}} = \Gamma$ on $\pi_1(V_{\bar{k}})$ be faithful, i.e.

$$\Gamma \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi)$$

m1

Marginal note. [m1] this homomorphism[m] is automatically factored through the subgroup $\mathcal{N}(\pi)$ which interests us [illeg. 1]|normalizes $\widehat{\mathcal{T}}(\pi)$

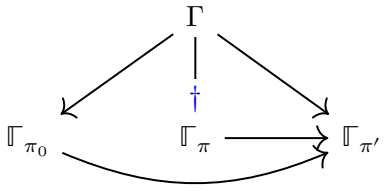
injective, but even the composite homomorphism

$$\Gamma \longrightarrow \mathcal{N}_4(\pi)|\mathcal{N}_\Delta(\pi) \longrightarrow \mathcal{N}(\pi)/\widehat{\mathcal{T}}^+(\pi) \simeq \mathbb{\Gamma}_\pi$$

should be injective (in which case it will even be an isomorphism, since it is surjective by the very construction of $\mathcal{N}(\pi), \mathbb{\Gamma}_\pi$…) Some heuris-

[scan 503]

tic arguments made previously (probably less convincing than those of the present § !) seem to indicate that this would be the case at least when V is an étale covering of $U = (U_{03})_k$, in which case indeed the homomorphism $\Gamma \longrightarrow \mathbb{\Gamma}_\pi$ fits into a commutative diagram [p. 255]



where $\pi_0 = \pi(U_{\bar{k}})$, so that π is an open subgroup of π_0, and $\pi''|\pi'$ is a suitable open subgroup, characteristic in π_0. If we could show that $\mathbb{\Gamma}_{\pi_0} \longrightarrow \mathbb{\Gamma}_{\pi'}$ is injective (which is $\pm$ a matter of profinite algebra), this would likewise be true of the composite $\Gamma \hookrightarrow | \subset \mathbb{\Gamma}_{\pi_0} \longrightarrow \mathbb{\Gamma}_{\pi'}$, thus also of $\Gamma \longrightarrow \mathbb{\Gamma}_{\pi}$. Thus modulo this hypothesis on profinite groups, and using Belyi, we find that

---

Uncertainty retained from cancelled passages.

140-2-s503-D001 $\longrightarrow$

for every algebraic curve V over k, $\exists\, V' \subset V$ nonempty open, such that [scan 504]

$$\Gamma \longrightarrow \mathbb{\Gamma}_{\pi'} \qquad (\text{where } \pi' = \pi_1(V'_{\bar{k}}))$$

is injective. As for knowing whether $\Gamma \longrightarrow \mathbb{\Gamma}_{\pi}$ is itself already injective – or † what amounts to the same thing, whether $\Gamma \longrightarrow \mathbb{\Gamma}_{g}$ for $g \geq 2$, and $\Gamma \longrightarrow \mathbb{\Gamma}_{1,1}$ are injective, I have no plausible heuristic reason to convince me of it at present – perhaps|may be this is entirely false ? The more or less convincing argument recalled previously (assuming we manage to justify it) would show only that if $\Gamma \longrightarrow \mathbb{\Gamma}_{\pi}$ is injective for one π (which depends only on its type (g, ν)) then it is so for the π' open in π. † We know it at

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Uncertainty retained from cancelled passages.

140-2-s504-D001 [illeg. 1]

140-2-s504-D002 it|-it

present (modulo very little, at any rate) for the types $(0, \nu)$ $(\nu \geq 3)$ exactly – no more, no less – and we could perhaps deduce it for types (g, ν) which are deduced from them by "finite covering". But it is already clear that we do not obtain the types $(g, 0)$ in this way $(g \geq 2)$, nor even any (g, ν) with $g \geq 1$, $\nu \in \{0, 1, 2\}$.|- That is to say we are far from the mark…The first really interesting case would thus be the case (1,1), where $\widehat{\mathcal{T}}^{+}_{1,1} \simeq$ Sl|SL$(2, \mathbb{Z})^{\wedge}$ Operating|operating by outer automorphisms on $\widehat{\Pi}_{1,1}$ (a free group on two generators, again – as if by chance) – is the "arithmetic" action of $\mathbb{\Gamma}_{\mathbb{Q}}$ on $\widehat{\Pi}_{1,1}$ (i.e. the outer action modulo $\widehat{\mathcal{T}}^{+}_{1,1} \simeq$ Sl|SL$(2, \mathbb{Z})^{\wedge}$) faithful ? [scan 505] [p. 256] torus with one hole !

36 The isomorphism $\mathbb{\Gamma}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{\Gamma}_{1,1}$, and the injectivity of $\mathbb{\Gamma}_{\mathbb{Q}} \longrightarrow \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\mathcal{T}}^{+}_{11}) \simeq \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}$(Sl|SL$(2, \mathbb{Z})^{\wedge}$) [m3] [scan 506] [p. 257]

Marginal note. [m3] The reflections|reflections of the present §, a bit halting, will be taken up again in a way|way less ponderous in the § that follows…

Let Π'_{03} be the quotient group of Π_{03} defined by the relations

$$(1) \qquad l_1^2 = l_\infty^3 = 1$$

– this is therefore the "oriented cartographic group for triangulated structures",|; the finite $\Pi'_{0,3}$-sets correspond|corresponding to oriented finite maps (possibly having flattened loops) whose faces are triangles or mono-angles. The outer operation of $\mathbb{\Gamma}_{\mathbb{Q}}$ (but not that of $\mathfrak{S}_3|\mathcal{S}_3$!) on $\widehat{\Pi}_{03}$ passes to the quotient as an action on $\widehat{\Pi}'_{0,3}$. We can improve theorem 1 of the preceding § by the|the

Proposition The outer action of $\mathbb{\Gamma}_{\mathbb{Q}}$ on $\widehat{\Pi}'_{0,3}$ is faithful.

To show this, we shall embed $\Pi_{0,3}$ as a subgroup of finite index in $\Pi'_{0,3}$, whence an analogous embedding †

$$(2) \qquad \widehat{\Pi}_{0,3} \hookrightarrow \widehat{\Pi}'_{0,3}$$

which will be compatible with the outer action of Γ.

Uncertainty retained from cancelled passages.
140-2-s506-D003 [illeg. 1]

[scan 507]

For this, consider the quotient scheme

$$Y = \underbrace{\mathbb{P}^1_{\mathbb{Q}}}_{X} / \mathfrak{S}_3$$

with the three points a_0, a_1, a_∞ of Y, rational over $\mathbb{Q}$, a_0 corresponding to the orbit $\{0, 1, \infty\}$, a_1 to the orbit $\{-1, \frac{1}{2}, 2\}$, and a_∞ to the orbit $\{-j, -\bar{j}\}$ of $\mathfrak{S}_3$ (which exhausts the set of geometric "singular" orbits). We know that Y is a projective line (over $\overline{\mathbb{Q}}$ a priori), which we pin by a_0, a_1, a_∞ as points $0, 1, \infty$, (thus $Y \simeq \mathbb{P}^1_{\mathbb{Q}}$ – the † morphism $X \longrightarrow Y$ is thus identified with a well-determined morphism $j = \exp \frac{2i\pi}{3}$

$$(3) \qquad f : \begin{array}{ccc} \mathbb{P}^1_{\mathbb{Q}} & \longrightarrow & \mathbb{P}^1_{\mathbb{Q}} \\ \| & & \| \\ X & & Y \end{array}$$

which makes X a Galois covering|covering of Y, with group $\mathfrak{S}_3$, étale over|over $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, with ramification indices at these points $2, 2, 3$. From the point of view of the geometry of maps (placing oneself|immersing oneself over the base field $\mathbb{C}$), we consider the

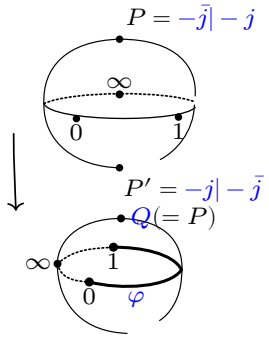
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Uncertainty retained from cancelled passages.
140-2-s507-D002 [illeg. 1]

[scan 508]

map determined on X by the spherical triangles $(0, 1, \infty)$ (on the real axis as equator) – which is thus the universal weighted triangular map – as the inverse image of the universal map (having a single vertex 0, a single edge (folded) $0 \longrightarrow 1$, a single face, with center ∞). m5 [p. 258 (259)]

Marginal note. m5



Every topological † étale covering of $U_{03}(\mathbb{C}) \subset X$ thus gives a topological étale covering of $U_{0,3}(\mathbb{C}) \subset Y$, having † over 1 ramification 2, over ∞ ramification 3 exactly|exactly, which corresponds to the "forgetful" functor (of the weighting), where a weighted triangular map is considered as just a map. From the point of view of fundamental groups we thus find an † equivalence of categories

†

$$(\dagger 4)\qquad \left(\text{Étale coverings of } X\setminus\{0,1,\infty\}=U_{0,3}\right)\ \simeq\ \left[\begin{array}{c}\text{Coverings of } Y\setminus\{0,1,\infty\}=U_{0,3}\\ \text{with ramification subordinate to}\\ \text{the signature } 2\{1\}+3\{\infty\}\end{array}\right]$$

$$\wr \qquad\qquad \Big/\text{the object } (X\setminus\{0,1,\infty\})\text{ of the said category}$$

$$\wr$$

$$\Pi_{0,3}\text{-sets}\qquad\qquad (\Pi'_{0,3}\text{-Sets})/\dagger E$$

where E is the $\Pi'_{0,3}$-set corresponding to the object $X\setminus\{0,1,\infty\}$ of the category of étale coverings of $Y\setminus\{0,1,\infty\}$ with ramification †subordin-ate to $2\{1\}+3\{\infty\}$

†

Uncertainty retained from cancelled passages.

140-2-s508-D001 of
140-2-s508-D003 [illeg. 1]
140-2-s508-D004 homomorphism
140-2-s508-D005 $X(\mathbb{C})$
140-2-s508-D008 [*illeg.* 1]
140-2-s508-D009 E
140-2-s508-D010 imposed
140-2-s508-D011 $\simeq\quad \Pi'^{\circ}_{0,3}$

[scan 509]

Here, † † the base point of $Y-\{0,1,\infty\}$ chosen;|, to describe the coverings † (with ramification subordinate to …) by a fundamental group with ramification, being again $P=-\bar{j}$, we shall have [p. 260|250]

$$(5)\qquad E=f^{-1}_{\mathbf{C}|\mathbb{C}}(P)$$

and we can thus make explicit the category $(\Pi'_{03}\text{-Sets})/E$;

Where E is a homogeneous space, isomorphic to the quotient of Π'_{03} by the subgroup|subgroup Π'°_{03} kernel of the suitable surjective homomorphism

$$(6)\qquad \left\{\begin{array}{rcl}\Pi'_{0,3} & \longrightarrow & \mathfrak{S}_3|\mathcal{S}_3\\ \ell_0 & \longmapsto & \text{element of order 2}\\ \ell_1 & \longmapsto & \text{element of order 2}\\ \ell_\infty & \longmapsto & \text{element of order 3}\end{array}\right\}\quad\text{to be computed !}$$

We choose a $Q_0\in E$ as "origin" of E – so as to be able to identify E as a $\Pi'_{0,3}$-set with $\Pi'_{03}/\Pi'^{\circ}_{03}\simeq\mathfrak{S}_3$ – then $(\mathrm{Ens}\,\Pi'_{03})/E$ is identified, by the functor $F\longmapsto F_{Q_0}$, with $\mathrm{Ens}(\Pi'^{\circ}_{03})$.

We thus find in summary an equivalence of categories (depending on the choice of Q_0)

$$\Pi_{03}\text{-sets}\quad\simeq\quad\Pi'^{\circ}_{03}\text{-sets}$$

which is itself described by a bitorsor under $(\Pi'^{\circ}_{0,3},\Pi_{03})$ – and, up to isomorphism (not unique !), by an isomorphism

$$\Pi_{03}\simeq\Pi'^{\circ}_{0,3}\quad -$$

the isomorphisms thus obtained (for

Uncertainty retained from cancelled passages.

140-2-s509-D001 [illeg. 3]

140-2-s509-D002 [illeg. 1]
140-2-s509-D003 f^{-1} (

[scan 510]
[p. 249]

varying origins of the bitorsor) form|forming exactly a conjugacy class under Π'°_{03} – i.e. an outer isomorphism $\Pi_{03} \simeq \Pi'^{\circ}_{03}$. When in addition the choice of Q_0 varies, we find a composite|composites

$$\Pi_{0,3} \longrightarrow \Pi'^{\circ}_{03} \longrightarrow \Pi'_{03}$$

† a Π'_{03}-conjugacy class, i.e. an outer homomorphism.

I have slaved away a lot for not much – for lack of having written down the necessary|elementary general functorialities for "fundamental groups with ramification". For example the mysterious bifunctor I from earlier is made up of classes of paths from $P \in X(\mathbb{C})|X_{\mathbb{C}} \setminus \{0,1,\infty\}$ (base point for computing $\Pi_{0,3}$) to Q_0 (playing the role of a new base point, having the merit of mapping to the one – P – which serves to compute Π'_{03}, fundamental group with ramification on $Y(\mathbb{C}) \setminus \{0,1,\infty\}$ …) It would perhaps be more convenient to define direct^{ly}|directly a functor in the opposite direction:|.

$$(7) \qquad \begin{array}{c}\text{Coverings of } Y(\mathbb{C})\setminus(0,1,\infty) \text{ subordinate}\\ \text{to the signature } 2\{1\}+3\{\infty\}\end{array} \longrightarrow \begin{array}{c}\text{Étale|Étale coverings of}\\ X(\mathbb{C}) \setminus \{0,1,\infty\}\end{array}$$

by inverse image, followed|followed by a normalization

Uncertainty retained from cancelled passages.
140-2-s510-D002 exactly